Micro

File: Micro.kicad_sch

Constant Current Supply

File: CCS.kicad_sch

IO

File: IO.kicad_sch

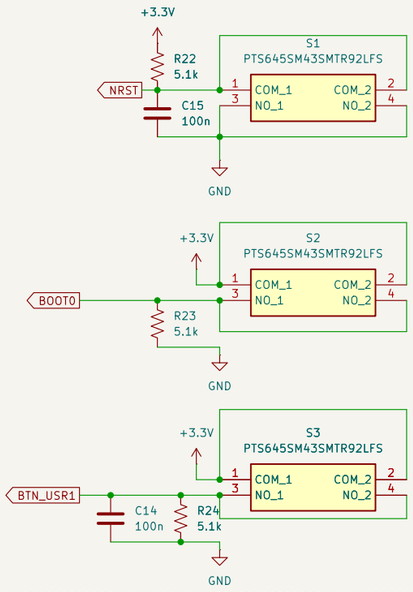
Cy Westbrook
Mark Tymoshenko mat399@scarletmail.rutgers.edu
Rutgers Capstone ECE 2026–2027

Sheet: /
File: Arcjet–PPU.kicad_sch

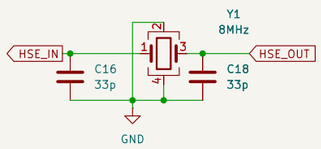
Title: Arc Jet PU (Power Unit)

Size: A4	Date: 2026–10–29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/4

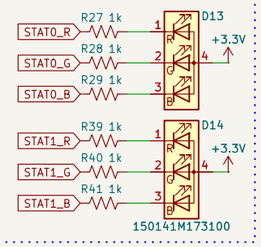
BUTTONS: RESET / BOOT0 / USER



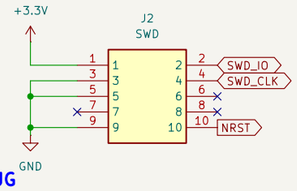
Do we actually need it?



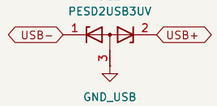
STATUS LEDS



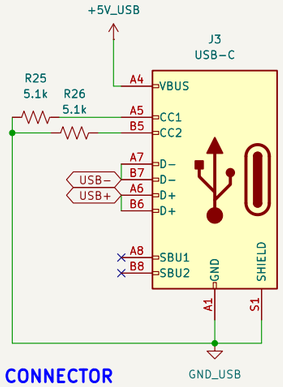
SWD DEBUG



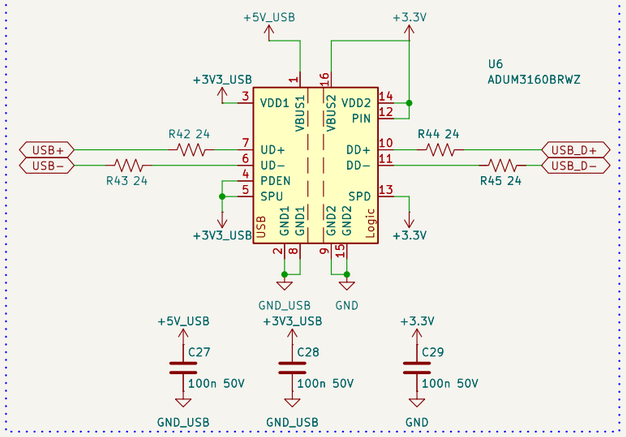
USB ESD



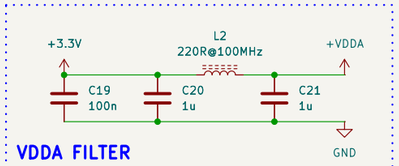
USB−C CONNECTOR



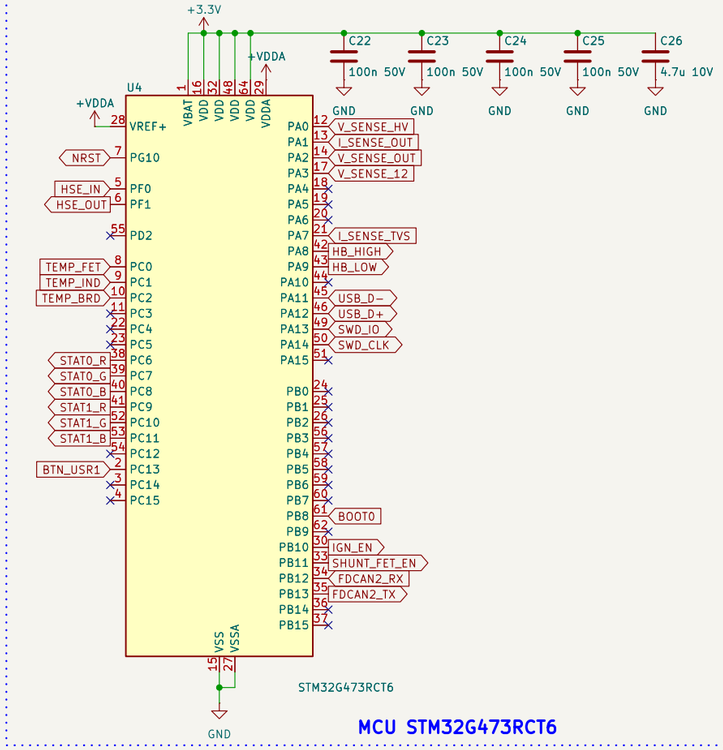
USB ISOLATION



VDDA FILTER



MCU STM32G473RCT6





Cy Westbrook
Mark Tymoshenko mat399@scarletmail.rutgers.edu

Rutgers Capstone ECE 2026–2027

Sheet: /Micro/
File: Micro.kicad_sch

Title: Arc Jet PU (Power Unit)

Size: A3	Date: 2026-10-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 7/4

DC INPUT, TVS & BULK CAPS

+DC_IN
PWR_FLAG
J4
DC_IN
R16 100k
V_SENSE_HV
R17 2.2k
C47 10n 50V
GND
C1 3.3u 250V
C2 3.3u 250V
C3 3.3u 250V
C4 3.3u 250V
C5 3.3u 250V
C6 3.3u 250V
SMCJ100CA
D9
GND

GATE DRIVER

+12V
C8 2.2u 25V
GND
U1
VDD HO
HI HB
LI HS
COM LO
UCC27712DR
HB_HIGH
HB_LOW
R1 3
D1 US1G
+12V
C9 2.2u 25V
R4 1
D3 1N4148WS
R5 3
10k R30
R8 1
R9 3
10k
1N4148WS
D5
Q1 IPB107N20N3G
Q3 IPB107N20N3G
R31

HALF BRIDGE, OUTPUT INDUCTOR & CURRENT SENSE

C7 3.3u 250V
C10 3.3u 250V
L1 100u 10.5A
R2 5m 1W
U3 INA281A2QDBVRQ1
IN+ IN−
GND OUT VS
+3.3V
C11 100n 50V
I_SENSE_OUT

OUTPUT VOLTAGE SENSE

R20 51k
R19 51k
R18 51k
R14 51k
+3.3V
D11 BAV199
R21 2.2k
V_SENSE_OUT
V_SENSE_OUT to non-FT ADC pin
R15 2.2k
C13 10n 50V
GND

ARC OUTPUT

J1
ARC_OUT
GND

OUTPUT PROTECTION & SHUNT FET

+DC_IN
D7 ES3J
C40 100n 250V
D2 SMCJ100CA
I_SENSE_TVS
R3 0.1 1W
+12V
C12 2.2u 25V
GND
U2 UCC27511ADBVR
OUTH VDD IN+
OUTL GND IN−
SHUNT_FET_EN
Q5 IPB107N20N3G
R12 3.3
R13 0
GND

TEMPERATURE SENSING

+3.3V
10k R36
TEMP_FET
C44 100n 50V
TH1 10k NTC
GND
10k R37
TEMP_IND
C45 100n 50V
TH2 10k NTC
10k R38
TEMP_BRD
C46 100n 50V
TH3 10k NTC

TH1 at Q1/Q3, TH2 at L1, TH3 near U1.

IGNITION DRIVE just an idea maybe will be offboard

+DC_IN
10k R32
10k R33
C41 3.3u 250V
C42 3.3u 250V
ES3J D8
J5
IGN_PRI
Ignition: C41/C42 discharge into off-board ignition transformer primary (J5) when IGN_EN pulses.
+12V
C43 1u 25V
U5
IN+ VDD OUTH
IN− GND OUTL
UCC27511ADBVR
R34 10
R63 0
R35 10k
Q6 IPB107N20N3G
IGN_EN
GND

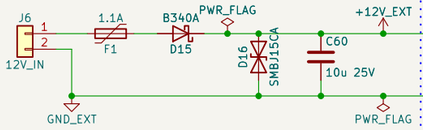
Cy Westbrook
Mark Tymoshenko mat399@scarletmail.rutgers.edu
Rutgers Capstone ECE 2026–2027

Sheet: /Constant Current Supply/
File: CCS.kicad_sch

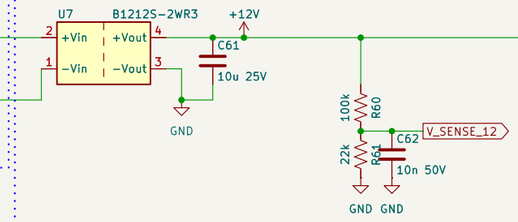
Title: Arc Jet PU (Power Unit)

Size: A3	Date: 2026-10-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 8/4

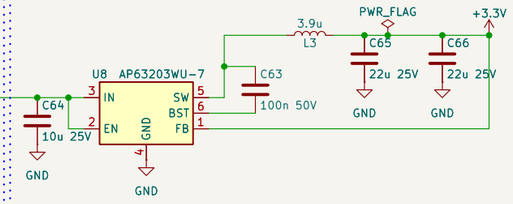
12V INPUT & PROTECTION



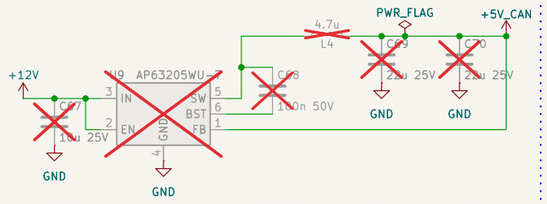
ISOLATED 12V & 12V MONITOR



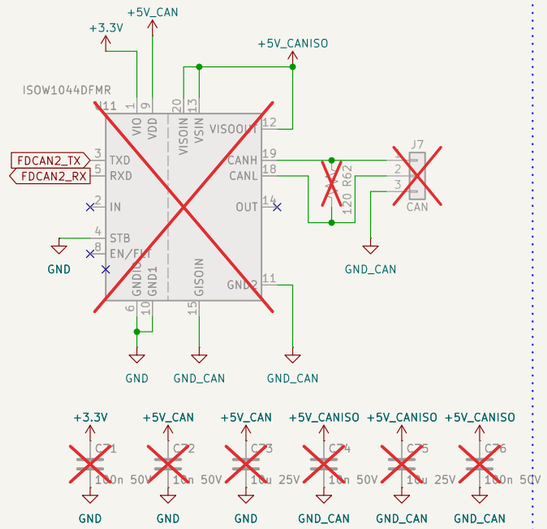
3.3V BUCK



CAN 5V SUPPLY (DNP)



ISOLATED CAN (DNP)



Cy Westbrook
Mark Tymoshenko mat399@scarletmail.rutgers.edu
Rutgers Capstone ECE 2026–2027

Sheet: /IO/
File: IO.kicad_sch

Title: Arc Jet PU (Power Unit)

Size: A3	Date: 2026-10-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 9/4